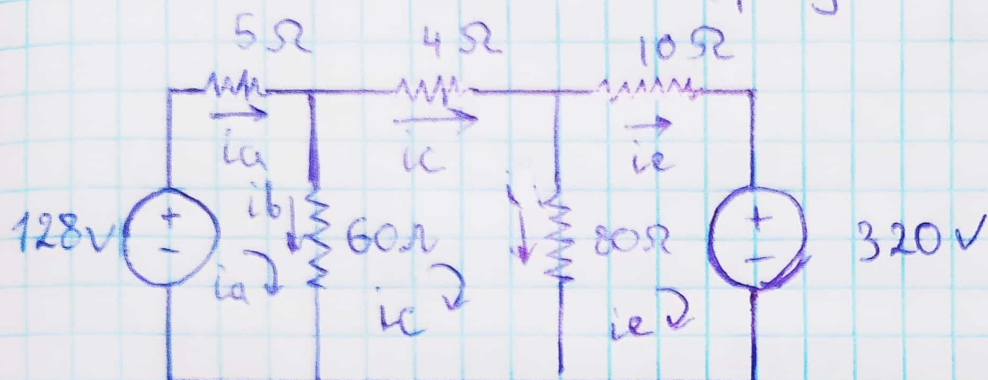


Homework #3

exercise 4.33 (page 134)



$$i_b = i_a - i_c$$

$$i_d = i_c - i_e$$

$$-128 + 5i_a + 60(i_a - i_c) = 0$$

$$4i_c + 80(i_c - i_e) + 60(i_c - i_a) = 0$$

$$320 + 80(i_e - i_c) + 10i_e = 0$$

$$i_a(5 + 60) + i_c(-60) + i_e \cdot 0 =$$

$$= i_a(-60) + i_e(4 + 80 + 60) + i_e(-80) =$$

$$= i_a(0) + i_c(-80) + i_e(80 + 10) = \underline{\underline{-320}}$$

$$\begin{cases} 65i_a - 60i_c = 128 \end{cases}$$

$$\begin{cases} 144i_c - 60i_a - 80i_e = 0 / :4 \end{cases}$$

$$\begin{cases} 90i_e - 80i_c = -320 / :10 \end{cases}$$

$$\begin{cases} 65i_a - 60i_c = 128 \end{cases}$$

$$\begin{cases} 36i_c - 15i_a - 20i_e = 0 \end{cases}$$

$$\begin{cases} 9i_e - 8i_c = -32 \end{cases}$$

$$i_a = -6,8 \text{ A}$$

$$i_c = -9,5 \text{ A}$$

$$i_e = -12 \text{ A}$$

$$i_b = i_a - i_c = 2,7 \text{ A}$$

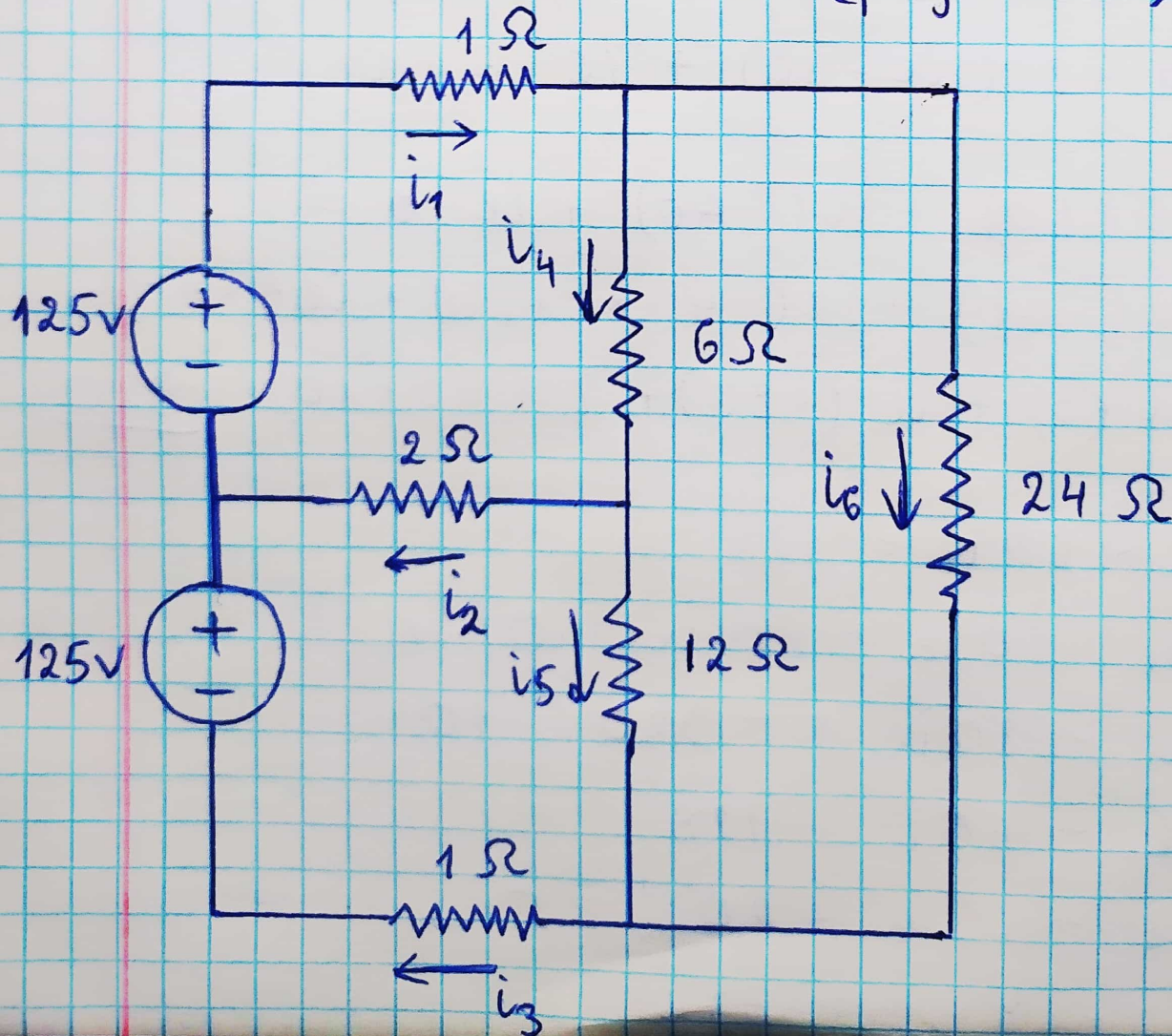
$$i_d = i_c - i_e = 2,5 \text{ A}$$

b) $p_{128\text{V}} = 870,4 \text{ W (abs)}$

$p_{320\text{V}} = -3840 \text{ W (dev)}$

total p developed: 3840 W

exercise 4.34 (page 134)



$$-125 + \dot{I}_1 + 6(\dot{I}_1 - \dot{I}_6) + 2(\dot{I}_1 - \dot{I}_3) = 0$$

$$-125 + 2(\dot{I}_3 - \dot{I}_1) + 12(\dot{I}_3 - \dot{I}_6) + \dot{I}_3 = 0$$

$$24 \dot{I}_6 + 12(\dot{I}_6 - \dot{I}_3) + 6(\dot{I}_6 - \dot{I}_1) = 0$$

$$9 \dot{I}_1 + (-2) \dot{I}_3 + (-6) \dot{I}_6 = 125$$

$$(-6) \dot{I}_1 + (-12) \dot{I}_3 + 42 \dot{I}_6 = 0$$

$$(-2) \dot{I}_1 + 15 \dot{I}_3 + (-12) \dot{I}_6 = 125$$

$$\dot{I}_2 = \dot{I}_1 - \dot{I}_3 = 5.33 \text{ A} \quad \left| \quad \dot{I}_1 = 23.76 \text{ A} \right.$$

$$\dot{I}_4 = \dot{I}_1 - \dot{I}_6 = 15.10 \text{ A} \quad \left| \quad \dot{I}_3 = 18.43 \text{ A} \right.$$

$$\dot{I}_5 = \dot{I}_3 - \dot{I}_6 = 9.77 \text{ A} \quad \left| \quad \dot{I}_6 = 8.66 \text{ A} \right.$$

$$P_s = P_t + P_b = -(425)(23.76) - (125)(18.43) \\ = -5273 \text{ W}$$

$$P_{1 \text{ top}} = (23.76)^2 \cdot 1 = 564.39 \text{ W}$$

$$P_2 = (5.33)^2 \cdot 2 = 56.79 \text{ W}$$

$$P_{1 \text{ bot}} = (18.43)^2 \cdot 1 = 339.59 \text{ W}$$

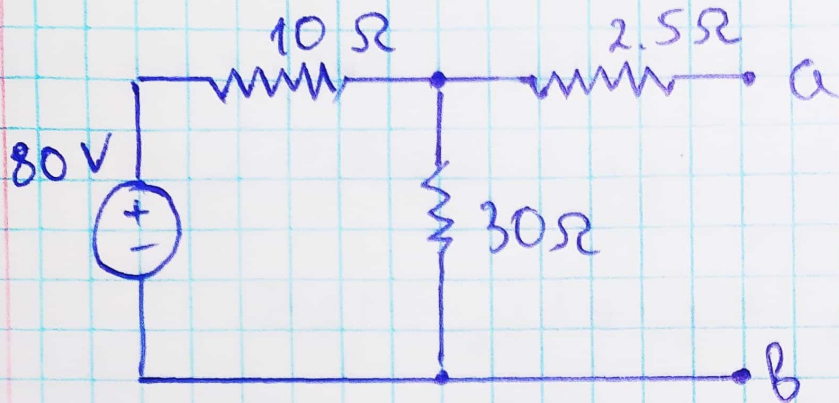
$$P_6 = (15.10)^2 \cdot 6 = 1367.64 \text{ W}$$

$$P_{12} = (9.77)^2 \cdot 12 = 1145.22 \text{ W}$$

$$P_{24} = (8.66)^2 \cdot 24 = 1789.47 \text{ W}$$

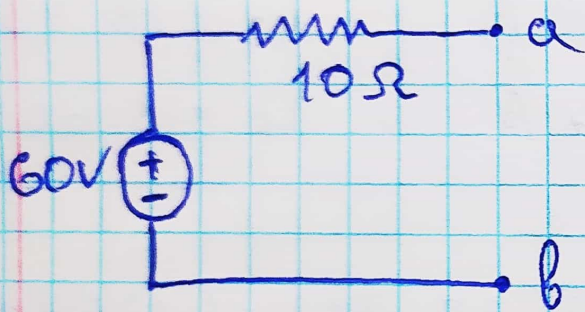
} 5273W

exercice 4.64 (page 137)

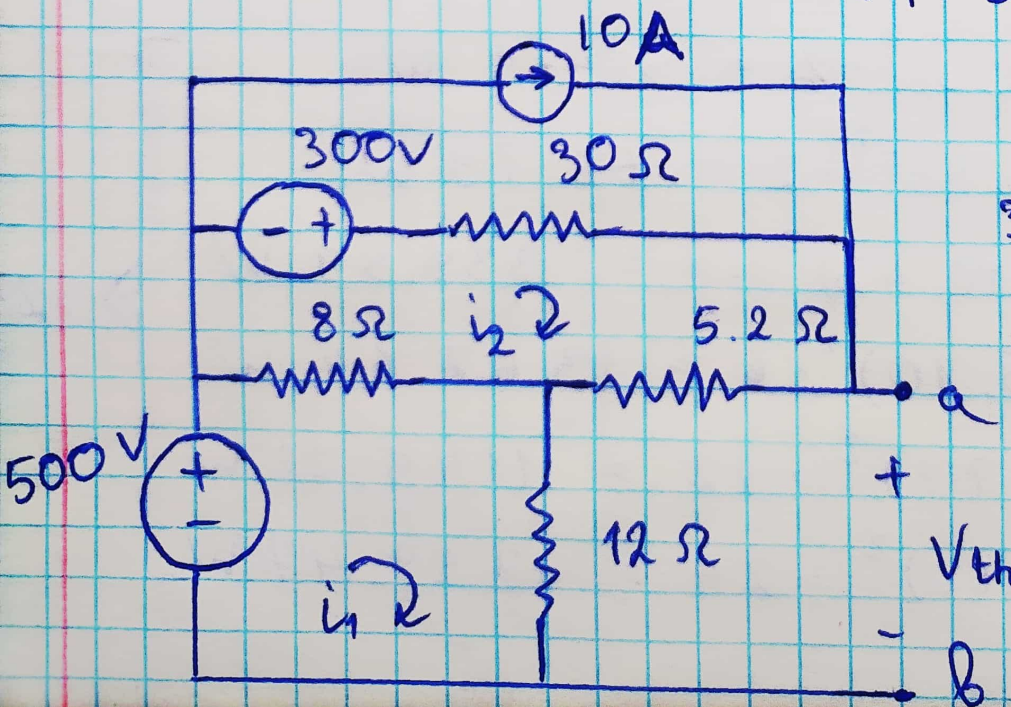


$$V_{th} = \frac{30}{40} \cdot 80 = 60V$$

$$R_{th} = 2.5 + \frac{30 \cdot 10}{40} = 10\Omega$$



exercice 4.67 (page 137)



$$500V = 20i_1 - 8i_2$$

$$300 = -8i_1 = 43.2i_2$$

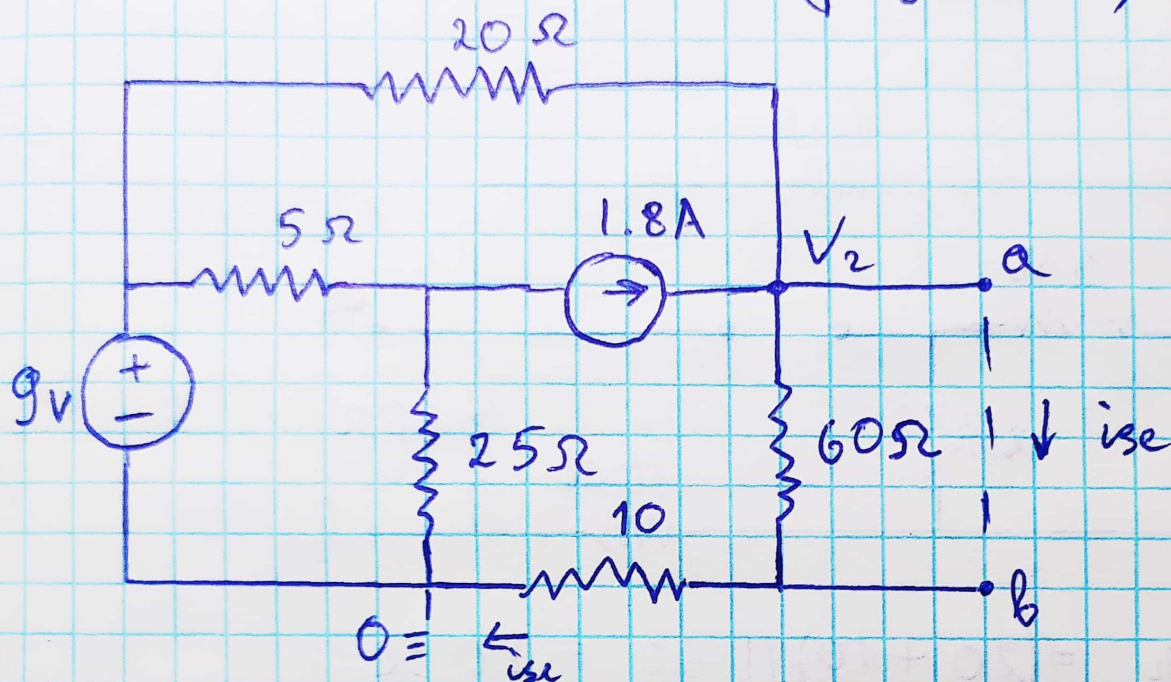
$$i_1 = 30A$$

$$i_2 = 12.5A$$

$$V_{th} = 12 i_1 + 5.2 i_2 = \underline{425 V}$$

$$R_{th} = (8 \parallel 12 + 5.2) \parallel 30 = \underline{7.5 \Omega}$$

exercice 4.78 (page 139)



Open circuit:

$$\frac{V_2 - 9}{20} + \frac{V_2}{70} - 1.8 = 0$$

$$V_2 = 35 V$$

$$V_{th} = \frac{60}{70} V_2 = 30 V$$

Short circuit:

$$\frac{V_2 - 9}{20} = \frac{V_2}{10} - 1.8 = 0$$

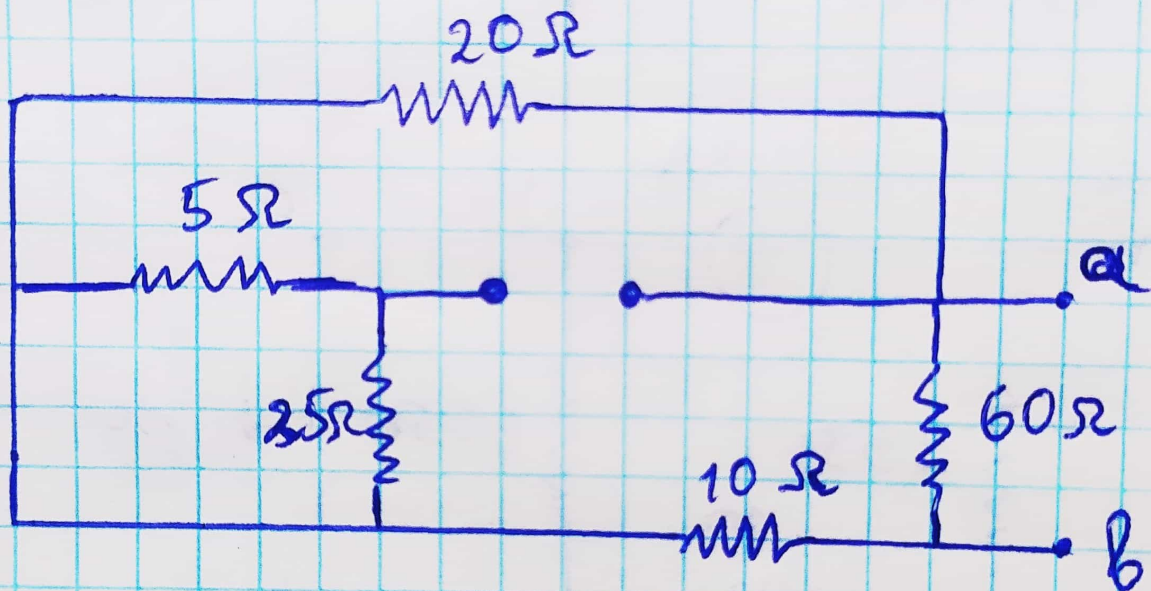
$$V_2 = 15 V$$

$$i_a = \frac{9 - 15}{20} = -0.3 \text{ A}$$

$$i_{sc} = 1.8 - 0.3 = 1.5 \text{ A}$$

$$R_{th} = \frac{30}{1.5} = 20 \Omega$$

b)



$$R_{th} = (20 + 10) \parallel 60 = 20 \Omega$$